

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Biological Sciences) Examination April 2018

Course code & Name: BS 304 Fundamentals of Microbiology II

Date: 03/04/2018 Day: Tuesday Time: 10:00 A M To 10:20 A M Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. In the following stage of binary fission, new cell wall material is synthesized and distributed equally in both the daughter cells
(i) mesosome attachment (ii) equatorial ridge formation
(iii) septum formation (iv) all of the above
2. Shrinking of a bacterial cell occurs in the following medium
(i) hypotonic (ii) acidic (iii) hypertonic (iv) alkaline
3. Chemostat is an example of the following culture:
(i) continuous (ii) Fed batch (iii) batch (iv) synchronous
4. The following test is most suitable for determination of the potency of disinfectants
(i) phenol coefficient (ii) E test (iii) disc diffusion (iv) all of the above
5. The following disinfectant can kill microorganisms by the generation of superoxide radicals
(i) ethanol (ii) formaldehyde (iii) glutaraldehyde (iv) peroxygens

6. Nutrient agar may be sterilized by _____.
(i) autoclaving (ii) filtration (iii) tyndalization (iv) hot air oven
7. The following group of microorganisms can grow above 80°C.
(i) thermophiles (ii) mesophiles (iii) hyperthermophiles (iv) all of the above
8. The following is (are) the indirect method(s) for determination of microbial growth
(i) turbidity (ii) protein content (iii) dry weight (iv) all of the above
9. The microflora that establish themselves briefly, but are excluded by competition from normal flora or by human immune system is called
(i) permanent (ii) transient (iii) resident (iv) all of the above
10. Biocontrol agent may act by following the number of pests in the field
(i) increasing (ii) equalizing (iii) decreasing (iv) all of the above

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Biological Sciences) Examination November 2017

Course code & Name: BS 304 Fundamentals of Microbiology II

Date: 03/04/2018 Day: Tuesday Time: 10:20 A M To 12:00 P M Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

- Q - II Answer the following questions as directed 10**
- A Write a short note on any two of the following (04)**
- i. Winogradsky's column with a labeled diagram
 - ii. Normal flora of human body
 - iii. Biofertilizer
- B Attempt any three of the following: (06)**
- i. Write a short note on: Probiotics
 - ii. Discuss briefly the methods used for sampling of air
 - iii. Discuss briefly: the microorganisms present in soil
 - iv. Write a short note on: Use of Peroxygens as disinfectant

SECTION - II

- Q-III Answer the following questions as directed 15**
- A Define any two the following terms (02)**
- i. Microbial death
 - ii. Generation time
 - iii. Synchronous culture
- B Attempt any two the following (04)**
- i. Classification of microorganisms on the basis of temperature requirement
 - ii. List the phases of bacterial growth and discuss lag phase in brief.
 - iii. Differentiate between batch culture and continuous culture
- C Suppose the generation time of a bacterium growing in exponential phase is 30 minutes, determine the number of generations after 6 hours (01)**
- OR
- C List the direct methods of measurement of growth**
- D Differentiate between any two the following (02)**
- i. Autoclaving and pasteurization
 - ii. Bactericidal and bacteriostatic agents
 - iii. Thermal death point and thermal death time
- E Attempt any three of the following (06)**
- i. Explain tyndelization in brief
 - ii. Write a short note on: Filtration as technique of sterilization
 - iii. List the methods of evaluating anti-microbial chemical agents